

CONCEPT 50.6

Skeletal systems transform muscle contraction into locomotion (pp. 1132–1136)

- Skeletal muscles, often in antagonistic pairs, contract and pull against the skeleton. Skeletons may be **hydrostatic** and maintained by fluid pressure, as in worms; hardened into **exoskeletons**, as in insects; or in the form of **endoskeletons**, as in vertebrates.
- Each form of **locomotion**—swimming, movement on land, or flying—presents a particular challenge. For example, swimmers need to overcome substantial friction, but face less of a challenge from gravity than do animals that move on land or fly.

? Explain how microscopic and macroscopic anchoring of muscle filaments enables you to bend your elbow.

TEST YOUR UNDERSTANDING

➔ For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

1. Which sensory receptor—category pair is correct?
(A) hair cell—nociceptor
(B) snake pit organ—mechanoreceptor
(C) taste receptor—chemoreceptor
(D) olfactory receptor—electromagnetic receptor
2. The middle ear converts
(A) air pressure waves to fluid pressure waves.
(B) air pressure waves to nerve impulses.
(C) fluid pressure waves to nerve impulses.
(D) pressure waves to hair cell movements.
3. During vertebrate skeletal muscle contraction, calcium ions
(A) break cross-bridges as a cofactor in hydrolysis of ATP.
(B) bind to troponin, exposing myosin-binding sites.
(C) transmit action potentials to the muscle fiber.
(D) spread action potentials through the T tubules.

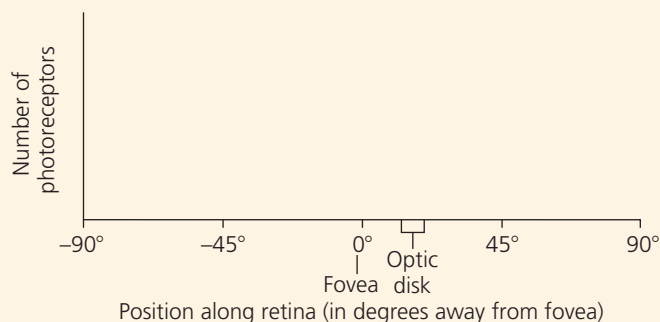
Levels 3-4: Applying/Analyzing

4. The human brain differentiates tastes from smells because action potentials for the two sensations differ in
(A) magnitude and shape.
(B) threshold potential.
(C) where they are received in the brain.
(D) how long they take to reach the brain.
5. The transduction of sound waves into action potentials occurs
(A) in the tectorial membrane as it is stimulated by hair cells.
(B) when hair cells are bent against the tectorial membrane, causing them to depolarize and release neurotransmitter that stimulates afferent neurons.
(C) as the basilar membrane vibrates at different frequencies in response to the varying volume of sounds.
(D) within the middle ear as the vibrations are amplified by the malleus, incus, and stapes.

Levels 5-6: Evaluating/Creating

6. Metal objects cause sharks to misdirect their bites. Sharks also can find batteries buried under sand. These facts suggest that sharks track their prey just before they bite in the same way that
(A) a rattlesnake finds a mouse in its burrow.
(B) an insect avoids being stepped on.
(C) a star-nosed mole locates its prey in tunnels.
(D) a platypus locates its prey in a muddy river.

7. **DRAW IT** Based on the information in the text, fill in the following graph. Use one line for rods and another line for cones.



8. **EVOLUTION CONNECTION** In general, locomotion on land will require more energy than locomotion in water. By integrating what you learned about animal form and function in Unit 7, discuss some of the evolutionary adaptations of mammals that support the high energy requirements for moving on land.
9. **SCIENTIFIC INQUIRY • INTERPRET THE DATA** To demonstrate how energy is stored in tendons during hopping, an instructor asked student volunteers to hop at a frequency that felt “natural” and then, after resting, to hop at exactly half that frequency. Hopping was done at a standard height and volunteers’ mass, O₂ consumption, and CO₂ production were measured. Here is a representative set of results calculated for one student.

Frequency (hops/sec)	Energy used (joules/sec)
1.85	735
0.92	716

The student consumed 159 joules/sec when standing. For each hop frequency, subtract this standing value from the energy used during hopping. Then divide by the hop frequency to calculate the energy cost per hop. How does the energy cost per hop differ at the two frequencies, and how might this be related to energy storage in tendons?

10. **WRITE ABOUT A THEME: ORGANIZATION** In a short essay (100–150 words), describe three ways in which the structure of the lens of the human eye is well adapted to its function in vision.
11. **SYNTHESIZE YOUR KNOWLEDGE**



Bloodhounds, which are adept at following a scent trail even days old, have no more olfactory receptor genes than other dogs. Predict how the sensory and nervous systems of bloodhounds differ from those of other dogs in ways that contribute to their tracking ability.

For selected answers, see Appendix A.